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FA23-BCE-113
Numerical Computation
Lab Assignment # 01

Q: Algorithm for modified Newton Raphson Method :

1. Standard vs Newton–Raphson for multiple roots

```
clc; clear; close all;
% Function and derivative
f = @(x) x.^3 - 2*x.^2 + 4*x - 8;
df = @(x) 3*x.^2 - 4*x + 4;
% Initial guess
x0 = 2.5;
tol = 1e-6;
maxIter = 10;
% simple method
fprintf('--- Newton–Raphson Method ---\n');
for i = 1:maxIter
x1 = x0 - f(x0)/df(x0);
fprintf('Iteration %2d: x = %.6f\n', i, x1);
if abs(x1 - x0) < tol
break;
end
x0 = x1;
end
fprintf('Root ≈ %.6f found in %d iterations\n', x1, i);
% modified method
m = 2; % multiplicity
fprintf('--- Modified Newton–Raphson Method ---\n');
for i = 1:maxIter
x1 = x0 - m * (f(x0)/df(x0));
fprintf('Iteration %2d: x = %.6f\n', i, x1);
if abs(x1 - x0) < tol
break;
end
x0 = x1;
end
fprintf('Root ≈ %.6f found in %d iterations\n', x1, i);
```

OUTPUT:

```
--- Newton–Raphson Method ---
Iteration 1: x = 2.098039
Iteration 2: x = 2.004576
Iteration 3: x = 2.000010
Iteration 4: x = 2.000000
Iteration 5: x = 2.000000
Root ≈ 2.000000 found in 5 iterations
--- Modified Newton–Raphson Method ---
Iteration 1: x = 2.000000
Root ≈ 2.000000 found in 1 iterations
>>
```

2. Standard vs Damped Newton–Raphson

```
clc; clear; close all;
f = @(x) x.^3 - 2*x + 2;
df = @(x) 3*x.^2 - 2;
x0 = 1.5;
tol = 1e-6;
maxIter = 30;
% --- Standard Newton–Raphson ---
fprintf('--- Standard Newton–Raphson ---\n');
x = x0;
for i = 1:maxIter
x1 = x - f(x)/df(x);
fprintf('Iteration %2d: x = %.6f\n', i, x1);
if abs(x1 - x) < tol
break;
end
x = x1;
end
fprintf('Standard Method Root ≈ %.6f found in %d iterations\n\n', x1,
i);
% --- Damped Newton–Raphson ---
alpha = 0.5; % damping factor
fprintf('--- Damped Newton–Raphson (α = %.2f) ---\n', alpha);
x = x0;
for i = 1:maxIter
x1 = x - alpha * (f(x)/df(x));
fprintf('Iteration %2d: x = %.6f\n', i, x1);
if abs(x1 - x) < tol
break;
end
x = x1;
end
fprintf('Damped Method Root ≈ %.6f found in %d iterations\n', x1, i);
```

OUTPUT:

```
--- Standard Newton–Raphson ---
Iteration 1: x = 1.000000
Iteration 2: x = 0.000000
Iteration 3: x = 1.000000
.
.
.
Iteration 29: x = 1.000000
Iteration 30: x = 0.000000
Standard Method Root ≈ 0.000000 found in 30 iterations
--- Damped Newton–Raphson (α = 0.50) ---
Iteration 1: x = 1.250000
Iteration 2: x = 0.979651
Iteration 3: x = 0.421791
.
.
.
Iteration 26: x = -1.769294
Iteration 27: x = -1.769293
Damped Method Root ≈ -1.769293 found in 27 iterations
>>
```